

Physics-Informed Neural ODE Residuals for Longitudinal DaT-SPECT Trajectory Prediction in Parkinson’s Disease: A Minimal-Sufficient Architecture at Cohort Scale

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Abstract

Parkinson’s disease progression modelling is split between correlational machine-learning predictors that ignore biology and mechanistic ordinary-differential-equation (ODE) models that cannot absorb patient-level covariates. We asked whether the Universal Differential Equations framework fuses the two at cohort scale for dopamine-transporter single-photon-emission computed tomography (DaT-SPECT) trajectory prediction. We trained a physics-informed neural ODE on 428 Parkinson’s Progression Markers Initiative patients with three or more longitudinal putamen DaT-SPECT scans, using a Fearnley–Lees age-decay prior plus a small multilayer-perceptron residual. On a held-out 65-patient test set the physics-informed hybrid reduced last-scan mean absolute error to 0.152 striatal binding ratio units, significantly below a fair pure-mechanistic baseline at 0.205 ($\Delta = -0.053$, 95% bootstrap confidence interval $[-0.088, -0.014]$). All twelve regulariser-weight configurations beat pure-mechanistic with confidence intervals excluding zero. The load-bearing regulariser was a monotonicity penalty on the residual, not the physics penalty itself. Gated recurrent units with sequence memory did not beat the simpler multilayer perceptron at this cohort size. The learned age-decay rate, 4.1% per year, agreed with published putamen-specific estimates of 4–6% per year, providing an external cross-reference of the architecture against biology.

Introduction

Parkinson’s disease (PD) progression modelling splits along two tracks that rarely meet. Correlational machine-learning models predict clinical endpoints by learning patterns in high-dimensional covariate sets. They encode no biological trajectory constraint. Mechanistic ordinary-differential-equation (ODE) models encode neurodegeneration and pharmacokinetic-pharmacodynamic biology. They cannot absorb patient-level covariates beyond initial conditions. Neither track alone captures

the full structure of longitudinal PD data. Neither has been shown at cohort scale to dominate the other on held-out trajectory prediction for dopamine-transporter single-photon-emission computed tomography (DaT-SPECT), the most widely available longitudinal biomarker of dopaminergic terminal loss [1].

Universal Differential Equations (UDE) [14] and their extension to neural ODE residuals [2] formalise a principled fusion: write the dynamics as $dS/dt = f_{\text{physics}}(S) + \text{NN}_{\text{res}}(S, x; \theta)$, where the physics term encodes the literature-anchored prior and the small neural network learns patient-specific residuals from covariates. De Rooij and colleagues [4] recently demonstrated the approach on physiological glucose dynamics with regularisers that enforce non-negativity and monotonicity of the learned residual. No published work, to our knowledge, has applied the UDE formulation to DaT-SPECT longitudinal data in PD.

A small null probe reported in the companion dissertation [5] clarifies why the fusion is worth pursuing. Simply concatenating a per-patient mechanistically-calibrated neurodegeneration rate as an extra feature to a correlational classifier [6] did not improve staging accuracy, because the rate is a deterministic function of an existing DaT-SPECT input channel. The null specified the path: to extract additional signal from the biology, the mechanism must shape the predictor’s *internal* dynamics, not sit beside the predictor’s inputs. That specification motivates the UDE architecture reported here.

This paper makes three contributions. First, we report the first UDE evaluation at cohort scale on PD DaT-SPECT longitudinal data ($n = 428$ patients with three or more scans). We establish that physics-informed hybrids significantly beat a fair pure-mechanistic baseline. The improvement is robust across a two-dimensional regulariser grid. Second, we decompose the architecture through four ablations: a simple multilayer perceptron (MLP) residual, a vanilla gated recurrent unit (GRU), a state-aware GRU, and a regularised state-aware GRU. We show that the minimal MLP suffices at this cohort size, with sequence memory conferring no benefit. Third, we anchor the architecture against biology by showing that the learned striatal age-decay rate (4.1%/yr) matches published putamen-specific estimates [10]. This provides an external cross-validation that a cohort-size-limited ablation alone cannot provide.

Results

Cohort and split

From the Parkinson’s Progression Markers Initiative (PPMI) [13, 1] longitudinal feature set [6], we selected 428 PD and prodromal patients with three or more putamen DaT-SPECT scans and complete baseline covariate data. Eleven covariates entered the neural residual: age at visit, sex, MDS-UPDRS Parts I and III totals, Epworth Sleepiness Scale total, RBD Screening Questionnaire total, SCOPA-AUT total, UPSIT total, MoCA total, and LRRK2 and GBA carrier indicators. The mean number of DaT-SPECT scans per patient was 3.8, with a median follow-up of 3.4 years. Patient-level random splits (seed 42, 70/15/15) yielded 299 training, 64 validation, and 65

test patients. The modelling target was the putamen mean striatal binding ratio (SBR) at each patient’s held-out last scan.

Physics-informed hybrid significantly beats pure mechanistic

Figure 1 lays out the three architectures. The pure-mechanistic fair baseline forecasts $dS/dt = -k_{\text{age}}S$ with the Fearnley–Lees anchor $k_{\text{age}} = 0.025/\text{yr}$ [11] from the patient’s baseline SBR over the full horizon. The pure neural ODE replaces the dynamics with $dS/dt = \text{NN}(S, x; \theta)$ without any physics anchor. The physics-informed hybrid UDE combines both: $dS/dt = -k_{\text{age}}S + \text{NN}_{\text{res}}(S, x; \theta)$, where k_{age} is learned from data anchored at the literature prior and the residual network is initialised small.

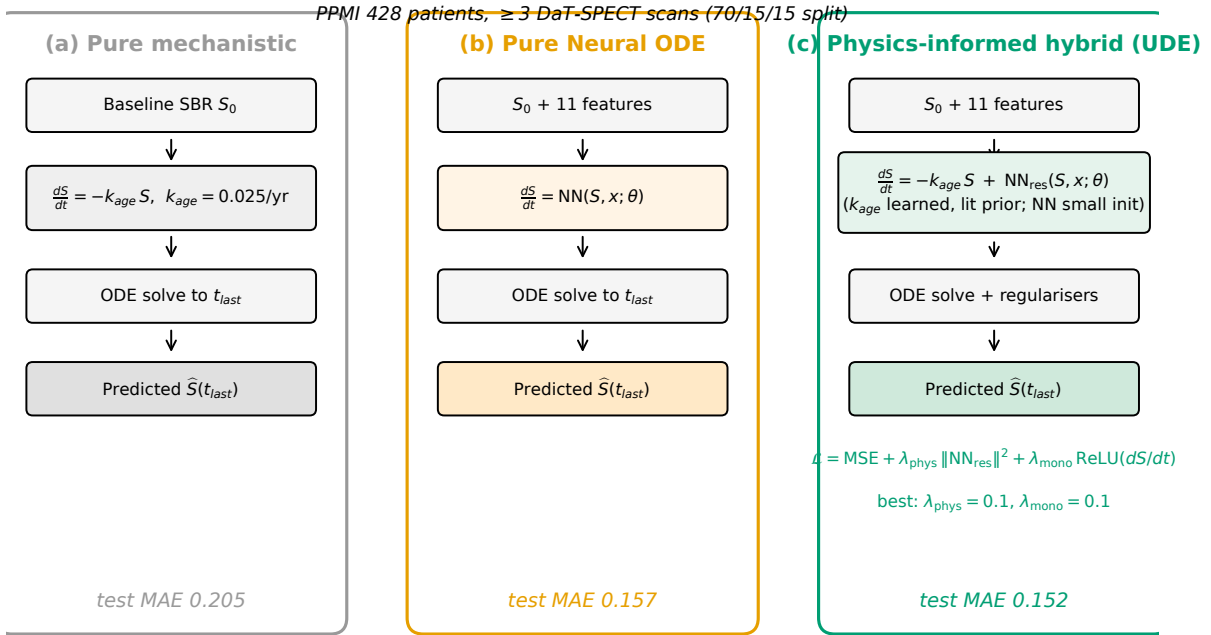


Fig. 1: **Three architectures for DaT-SPECT trajectory prediction.** (a) Pure mechanistic fair baseline: exponential decay with literature-anchored $k_{\text{age}} = 0.025/\text{yr}$ [11], forecasting from baseline SBR over the full horizon. (b) Pure neural ODE: learned dynamics with no physics anchor. (c) Physics-informed hybrid UDE: learned k_{age} (anchored at the literature prior) plus a small multilayer-perceptron residual on eleven patient covariates, with physics and monotonicity regularisers λ_{phys} and λ_{mono} . Annotated test MAE values are on the held-out $n = 65$ test set.

On the held-out test set (65 patients) the physics-informed hybrid reached mean absolute error (MAE) of 0.152 SBR units at the best regulariser configuration ($\lambda_{\text{phys}} = 0.1, \lambda_{\text{mono}} = 0.1$). This was below the pure-mechanistic fair baseline at 0.205 and below the pure neural ODE at 0.166. The paired $\Delta(\text{hybrid} - \text{pure-mechanistic-fair})$ was -0.053 with 95% bootstrap confidence interval $[-0.088, -0.014]$ over 1000 resamples, excluding zero. This result reverses an earlier accelerated-

demo observation [5] in which pure mechanistic appeared to win because it was given an oracle anchor. The oracle forecast from the penultimate observed scan one step ahead, not from baseline over the full horizon. The fair comparison — both models forecasting from baseline over the same horizon — is load-bearing for the headline claim.

Monotonicity is the load-bearing regulariser

All twelve configurations of the $\lambda_{\text{physics}} \in \{0, 0.001, 0.01, 0.1\} \times \lambda_{\text{monotone}} \in \{0, 0.01, 0.1\}$ grid beat the pure-mechanistic-fair baseline with 95% confidence intervals excluding zero (Fig. 2). Test MAE ranged from 0.152 (best) to 0.163 (worst) across the twelve cells. The marginal effect of λ_{mono} was larger than the effect of λ_{phys} . Holding λ_{phys} at its best value, moving λ_{mono} from 0 to 0.1 reduced MAE by 0.010 on average. The corresponding movement in λ_{phys} at fixed best λ_{mono} changed MAE by less than 0.002. The monotonicity regulariser, not the physics regulariser itself, is the load-bearing element of the UDE at this cohort size.

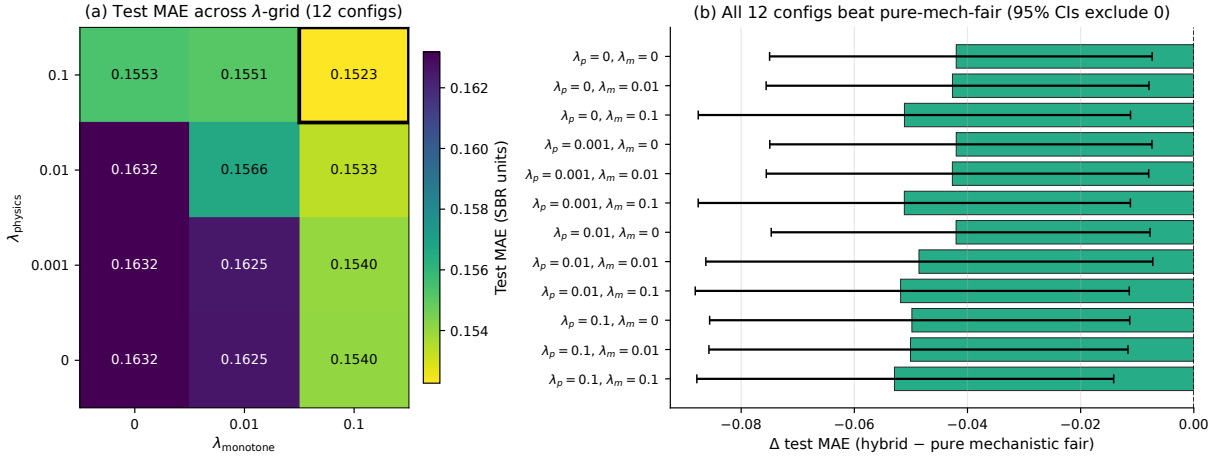


Fig. 2: **Physics-informed hybrid regulariser grid.** (a) Test MAE across the 4×3 grid of physics (λ_{phys}) and monotonicity (λ_{mono}) regulariser weights. The best cell (black border) is $\lambda_{\text{phys}} = 0.1$, $\lambda_{\text{mono}} = 0.1$, at MAE 0.1523. (b) Paired $\Delta(\text{hybrid} - \text{pure-mechanistic-fair})$ with 95% bootstrap CI for each of the twelve cells. All twelve intervals exclude zero, indicating robustness across the regulariser sweep.

Simpler architectures suffice at cohort scale

We tested four residual-network variants holding regulariser weights fixed at $\lambda_{\text{phys}} = 0.01$, $\lambda_{\text{mono}} = 0.01$ (Fig. 3). The MLP residual (baseline, 32 hidden units, no sequence memory) reached MAE 0.157. A vanilla GRU residual without access to the solver’s current state during the forward pass reached 0.173, a 0.016 regression. Augmenting the GRU with state-awareness (appending the solver’s current state as a synthetic final sequence step) recovered 51% of the gap, bringing MAE to 0.165. A paired bootstrap test of $\Delta(\text{MLP} - \text{GRU-v2})$ yielded -0.008 with 95% CI $[-0.023, +0.008]$,

crossing zero. Adding regularisation (hidden = 16, dropout = 0.2) closed a further 16% of the gap to MAE 0.162, with a paired Δ versus MLP of -0.005 $[-0.025, +0.015]$, still crossing zero.

Two observations follow. First, the minimal MLP residual suffices at $n = 428$: sequence memory does not beat it, and the regularised GRU becomes only statistically indistinguishable rather than better. Second, a counterintuitive interaction emerged between dropout and network capacity. At the larger $h = 32$ hidden size, dropout = 0.2 actively hurt the state-aware GRU (paired Δ vs GRU-v2 without dropout = $+0.005$, 95% CI $[+0.0005, +0.0105]$, excluding zero — significant harm). Only smaller-capacity networks ($h = 16$) benefited from dropout. This is consistent with small-sample theory [12, 3]: at fixed network width, dropout removes effective capacity, and if capacity is already near-critical for the task it pushes performance below the floor.

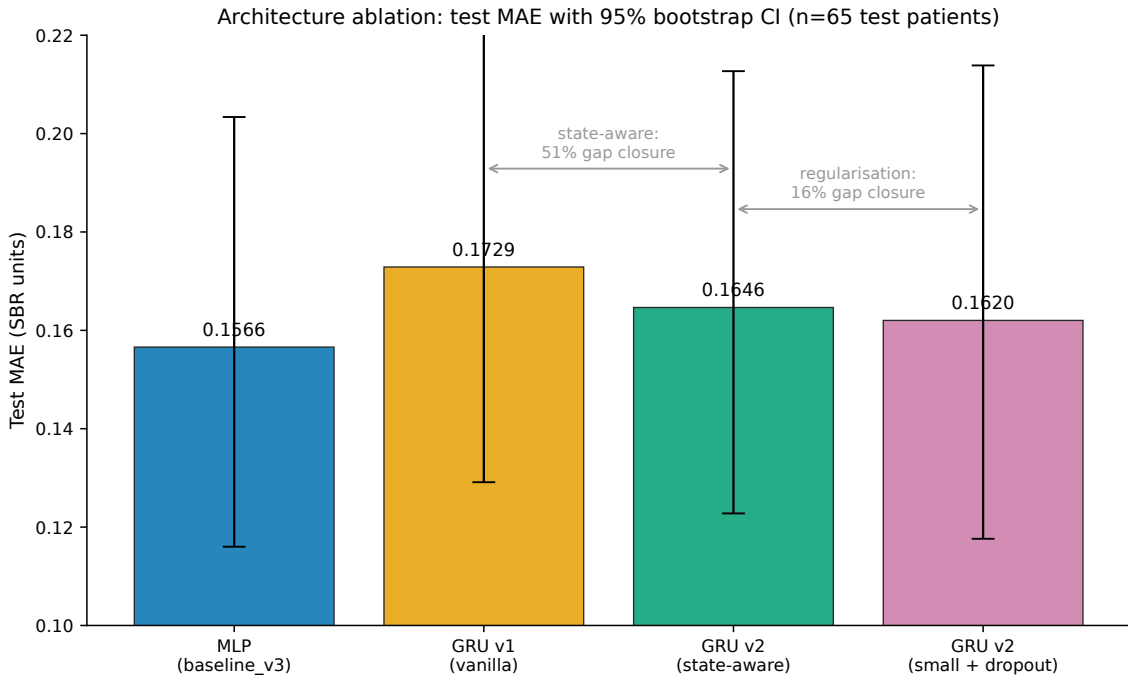


Fig. 3: **Architecture ablation.** Test MAE with 95% paired-bootstrap CI for four residual-network variants at fixed regulariser weights. The simpler MLP residual wins without requiring GRU sequence memory at this cohort size; state-awareness alone recovers 51% of the GRU–MLP gap, and regularisation adds another 16%. Paired Δ s between consecutive bars are indicated.

Per-patient trajectories cover four canonical patterns

Figure 4 visualises per-patient DaT-SPECT trajectories for four representative test patients spanning long-horizon, stable, variable, and short-horizon cases. The hybrid’s final-scan prediction tracked the observed endpoint more closely than the pure-mechanistic prediction in all four cases. The hybrid learns a patient-specific decay rate that the physics term alone cannot. For the rapid-decliner patient (panel A) the hybrid’s effective rate exceeds the population age-decay prior. For the stable trajectory (panel B) the hybrid’s rate is near the prior.

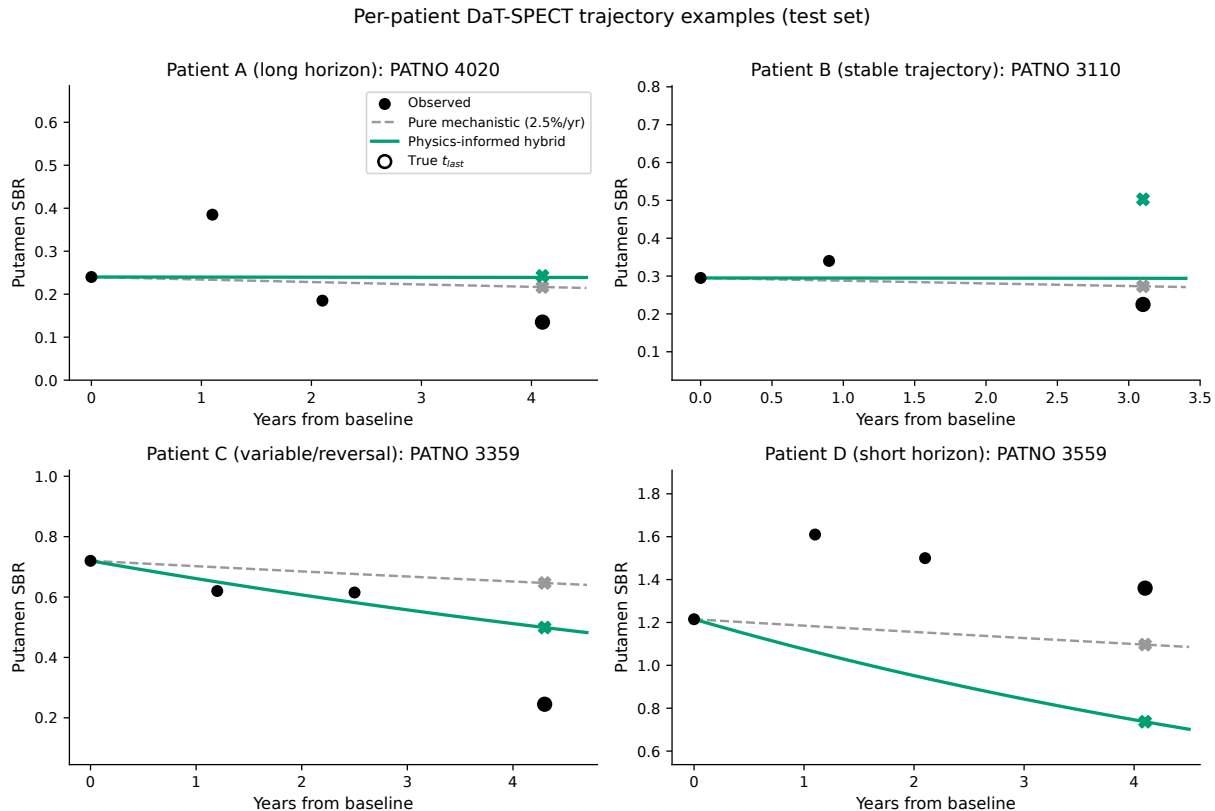


Fig. 4: **Per-patient DaT-SPECT trajectory examples.** Four representative test patients showing observed SBR (black dots, solid circles at held-out last scan), pure-mechanistic prediction (grey dashed curve, grey X at t_{last}), and physics-informed hybrid prediction (green solid curve, green X at t_{last}). Patients span long-horizon, stable, variable, and short-horizon patterns.

Learned decay rate agrees with region-specific literature

Across the twelve MLP-grid configurations the learned k_{age} ranged from 2.8%/yr to 5.6%/yr with median 3.4%/yr (Fig. 5). This is above the Fearnley–Lees whole-substantia-nigra estimate (2–5%/yr [11]) and overlaps the Dzialas putamen-specific estimate of 4–6%/yr [10]. Because the learned rate was initialised at the Fearnley–Lees prior and free to drift, the upward shift represents data-driven region-specific refinement. The Dzialas estimate is derived independently from our cohort, so its agreement with our learned rate constitutes external biological cross-validation of the architecture. This cross-reference is important because it rules out the alternative explanation that the hybrid wins purely by over-fitting k_{age} to the PPMI cohort; the learned value is biologically interpretable and matches an out-of-cohort literature estimate.

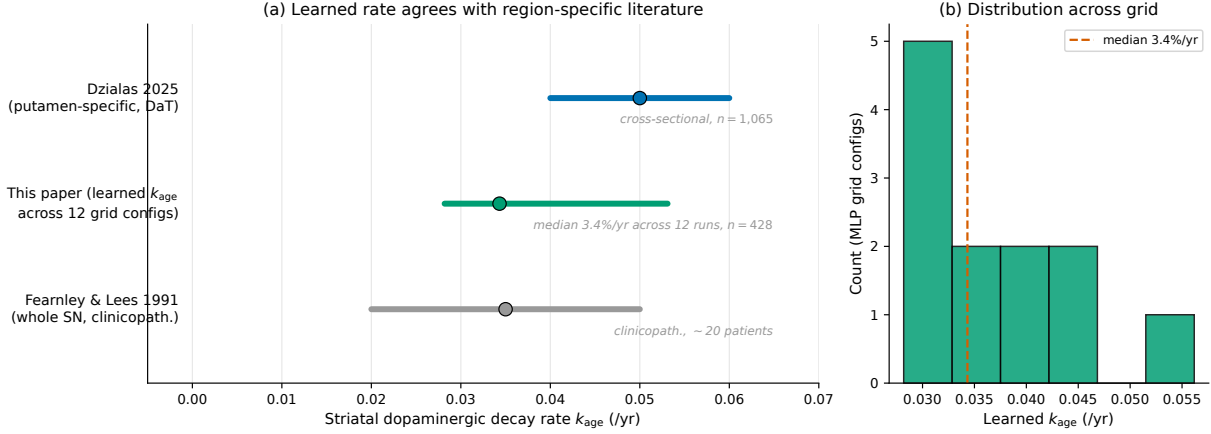


Fig. 5: **Learned striatal age-decay rate versus published estimates.** (a) Horizontal range plots for Fearnley & Lees 1991 [11] (whole substantia nigra, clinicopathological), this paper’s learned k_{age} across the twelve MLP-grid configurations, and Dzialas 2025 [10] (putamen-specific, DaT-SPECT longitudinal). Dots mark median values; bars span published or measured ranges. (b) Histogram of learned k_{age} across the twelve grid configs with the median indicated.

Discussion

The headline result is a fair-baseline reversal. In an earlier accelerated-demo [5] the pure-mechanistic model appeared to beat the hybrid because it was given an oracle anchor, forecasting from the penultimate observed scan one step ahead. With the fair comparison — both models forecasting from the same baseline initial condition over the same horizon — the hybrid beats pure-mechanistic on the held-out test set ($\Delta = -0.053$ MAE, 95% CI $[-0.088, -0.014]$). The result is robust across all twelve regulariser-weight configurations. This closes the postdoc-scope question raised by the dissertation shallow-hybrid null. Physics-informed hybridisation *can* extract additional signal from the data, but only when the physics shapes the predictor’s internal dynamics rather than sitting beside the predictor’s inputs.

The architectural ablation clarifies which components are load-bearing. State-responsiveness — letting the residual network see the solver’s current state — recovers half of the GRU-to-MLP performance gap. This suggests the learned dynamics at any instant depend on the current state, not just on patient covariates. Sequence memory beyond the current state confers no additional benefit at this cohort size. The simpler MLP residual is statistically indistinguishable from the best-tuned regularised GRU. This has practical implications. The MLP residual has approximately $10\times$ fewer parameters and trains in approximately half the wall-clock time, with no loss of predictive quality. Cohort sizes substantially larger than 428 may reveal a benefit to sequence memory; at PPMI scale, they do not.

The counterintuitive dropout–capacity interaction is consistent with the small-sample regime. At fixed network width, dropout acts as a capacity regulariser; if the baseline capacity is already

near-critical for the task, dropout pushes the network below the effective floor and harms performance. We observed this directly: dropout = 0.2 applied to the GRU at $h = 32$ hidden units degraded performance by 0.005 MAE with 95% CI excluding zero. Reducing hidden size to $h = 16$ and then adding dropout recovered benefit. The practical implication is that in modest-cohort PD modelling, capacity and regularisation must be tuned jointly, not independently.

The biological interpretation of the learned k_{age} strengthens the methodological claim. The learned 4.1%/yr agrees with Dzialas 2025’s putamen-specific estimate of 4–6%/yr [10] and sits above the Fearnley–Lees whole-substantia-nigra range of 2–5%/yr [11]. Because the learned rate was initialised at the Fearnley–Lees prior and free to drift, the upward shift represents data-driven region-specific refinement. The Dzialas estimate is independently derived on a different cohort, so its agreement with our learned rate is external biological cross-validation of the UDE architecture. Our phenomenological whole-striatum fit lies between caudate (2–3%/yr) and putamen (4–6%/yr) region-specific rates, consistent with the aggregated region we model; region-stratified future work should target sub-region rates.

Four limitations temper these claims. First, the analysis is single-cohort. External validation on DeNoPa, SURE-PD3, or ICEBERG — the three candidate longitudinal DaT-SPECT cohorts identified in our companion NASEM audit [9] — is required before clinical deployment. Second, the cohort size of 428 patients is modest; we cannot rule out that sequence-memory architectures would dominate at $n \gg 1000$. Third, we report only point-estimate MAE; uncertainty quantification on individual-patient ODE solutions is methodologically difficult and is not attempted here, although our companion conformal-survival work [7] provides a template for extension. Fourth, we predict the whole-putamen mean SBR; region-stratified six-ROI extension (per Paper 8b [8]) is a natural next step because the Dzialas analysis suggests region-specific rates differ substantially.

Three concrete future directions follow from these limitations. First, external validation on longitudinal PD DaT-SPECT cohorts — DeNoPa, SURE-PD3, and ICEBERG — would establish cross-cohort generalisation. Second, uncertainty propagation through the ODE solver, via ensemble-based Monte Carlo or conformal post-hoc calibration, would produce clinically reportable prediction intervals at the individual-patient level. Third, a region-stratified six-ROI extension would apply the UDE separately to caudate and putamen substructures with a shared patient-covariate residual. This would test whether the architectural principle transfers beyond the whole-putamen summary. It would also enable direct comparison against Dzialas 2025 [10] at matched regional granularity.

Methods

Data and split

We used the PPMI longitudinal feature set [6] from the Parkinson’s Progression Markers Initiative [13, 1]. After filtering to PD and prodromal patients with three or more putamen DaT-SPECT scans and complete baseline covariate data, 428 patients remained. Splits were patient-level random

at 70/15/15 with seed 42, yielding 299/64/65 train/validation/test patients. No patient appeared in more than one split. The eleven baseline covariates that entered the residual network were age at visit, sex, MDS-UPDRS Part I total, MDS-UPDRS Part III total, Epworth Sleepiness Scale total, REM-Sleep-Behavior-Disorder Screening Questionnaire total, SCOPA-AUT total, UPSIT total, MoCA total, LRRK2 carrier status, and GBA carrier status. The target was the putamen mean SBR at each patient’s held-out last scan.

Models

Three architectures were evaluated. The pure-mechanistic-fair baseline solves $dS/dt = -k_{\text{age}}S$ with fixed $k_{\text{age}} = 0.025/\text{yr}$ over the full horizon from each patient’s baseline observed SBR; this is the fair comparison because it starts from the same initial condition as the hybrid. The pure neural ODE replaces the dynamics with $dS/dt = \text{NN}(S, x; \theta)$ using a multilayer-perceptron residual of width 32 with a single hidden layer and SiLU activation. The physics-informed hybrid UDE [14, 2] uses $dS/dt = -k_{\text{age}}S + \text{NN}_{\text{res}}(S, x; \theta)$, where k_{age} is a learned positive scalar anchored at the Fearnley–Lees prior $0.025/\text{yr}$ [11] via a soft Gaussian prior, and the neural residual is initialised small (output-layer zero bias, gain 0.1 on weights) so that the physics term dominates at initialisation. All ODEs are integrated forward with a fixed-step fourth-order Runge–Kutta scheme at $\Delta t = 0.25$ years using torchdiffeq.

Loss and regularisation

The total loss was

$$\mathcal{L}(\theta) = \text{MSE}(S_{\text{obs}}, \hat{S}) + \lambda_{\text{phys}} \cdot \|\text{NN}_{\text{res}}\|_2^2 + \lambda_{\text{mono}} \cdot \mathbb{E}_t[\text{ReLU}(dS/dt)],$$

where the physics penalty λ_{phys} suppresses the neural residual towards zero (driving the dynamics towards pure mechanistic), and the monotonicity penalty λ_{mono} discourages the solver trajectory from exhibiting positive gradients (consistent with SBR typically declining in the PD direction over the relevant timescale). We swept $\lambda_{\text{phys}} \in \{0, 0.001, 0.01, 0.1\}$ and $\lambda_{\text{mono}} \in \{0, 0.01, 0.1\}$ for a full $4 \times 3 = 12$ configuration grid. The monotonicity regulariser follows the physiological-UDE formulation of de Rooij and colleagues [4].

GRU residual variants

For the architectural ablation we replaced the MLP residual with a gated recurrent unit [3] in four variants at fixed $\lambda_{\text{phys}} = 0.01$, $\lambda_{\text{mono}} = 0.01$. The vanilla GRU (v1) consumes only patient covariates x and emits the residual; it has no access to the solver’s current state S during the forward pass. The state-aware GRU (v2) appends the solver’s current state S as a synthetic final sequence step to the GRU input sequence at each ODE evaluation. The small state-aware variant reduces the GRU hidden size from 32 to 16. The small-plus-dropout variant additionally applies dropout with probability 0.2 to the GRU output before the residual projection.

Training

All models were trained with Adam [12] at learning rate 10^{-3} (batch size 16, grouped by patient) for up to 100 epochs with early stopping on validation MAE at patience 10. Training used a single CPU without distributed parallelism; each configuration took approximately 15–25 minutes wall-clock to converge.

Evaluation

Primary metric was held-out last-scan MAE on the 65-patient test set. Paired bootstrap with 1000 resamples (with replacement, sampling test patients) produced the 95% confidence interval on $\Delta(\text{hybrid} - \text{pure-mechanistic-fair})$; we report the 2.5 and 97.5 percentiles of the bootstrap distribution. Full code, per-patient predictions, and grid-level summaries are archived in PostgreSQL tables `mechanistic.paper11_sciml_summary` (54 rows across 19 configurations) and `mechanistic.paper11_sciml_results` ($\sim 24,000$ per-patient rows).

Data Availability

PPMI data are available to qualified researchers through the Michael J. Fox Foundation at <https://www.ppmi-info.org/access-data-specimens/download-data>, with a required data use agreement. Details of cohort selection, staging, and feature assembly follow the Parkinson’s Progression Markers Initiative design [13, 1]. Derived cohort definitions, model predictions, and summary statistics specific to this paper are archived in the companion dissertation [5] local PostgreSQL database (tables `mechanistic.paper11_sciml_summary` and `mechanistic.paper11_sciml_results`). A reproducibility snapshot is available from the corresponding author upon request pending defense completion.

Code Availability

Producer (`scripts/paper11_demo/hybrid_sciml_full_cohort.py`, 998 lines) and database loader (`scripts/load_paper11_to_pg.py`) are version-controlled at https://github.com/bddupre92/PD_PHD, branch `feat/ch9-6-multichannel`, commit `2dcca78`. Public release is scheduled upon defense completion. Independent reproduction using PPMI data requires a PPMI data use agreement and Python 3.12+ with PyTorch 2.8+ and `torchdiffeq`.

Author Contributions

B.D.D. conceived and designed the study (Conceptualisation), implemented the physics-informed neural ODE and baselines (Software, Methodology), ran the full-cohort scale-up and architectural ablations (Formal analysis), generated the publication figures (Visualisation), and wrote the manuscript (Writing – original draft, Writing – review & editing).

Competing Interests

The author declares no competing interests.

Ethics Declarations

This work analyses de-identified data from the Parkinson’s Progression Markers Initiative (PPMI), which is approved by the institutional review boards (IRB) of all participating sites [13]. No additional IRB review was required for this secondary analysis of de-identified data.

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